

Racketlon Match Prediction with Dynamic Ratings, Boosted Trees, and Geometric Confidence Analysis

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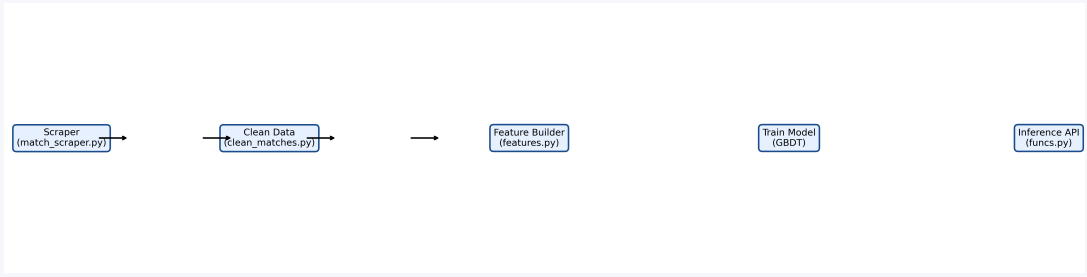
Motivation and Problem

Racketlon combines four sports—table tennis (TT), badminton (BD), squash (SQ), and tennis (TN)—into one total-points match.

This project built a full pipeline to:

- scrape historical match data,
- clean and normalize records,
- build leakage-safe pre-match features,
- predict sport scores and match winner,
- study confidence using KD-tree geometry.

Pipeline Overview



Stages

- scrape tournament data
- clean invalid matches
- build chronological features
- train sport-level regressors
- combine sport predictions
- analyze confidence

Data and Cleaning

Historical results were scraped from Tournament Software / FIR pages using a hybrid parser for both new and legacy formats.

Removed rows:

- doubles, mixed, team, and league events
- incomplete matches
- invalid or missing dates
- rows missing required scores

Players were normalized into canonical lowercase names.

Feature Engineering

Features were built chronologically, so each row only uses information available before the match.

Main feature groups:

- sport-specific rating differences
- recent rolling form
- residual performance vs expectation
- long-term shrunk means
- head-to-head statistics
- inactivity and time multipliers

Dynamic Rating Model

Expected score difference:

$$\hat{d} = 21 \tanh\left(\frac{\alpha(R_{p1} - R_{p2})}{2}\right)$$

with:

$$\alpha_{TT} = 0.060, \quad \alpha_{BD} = 0.080, \quad \alpha_{SQ} = 0.060, \quad \alpha_{TN} = 0.040$$

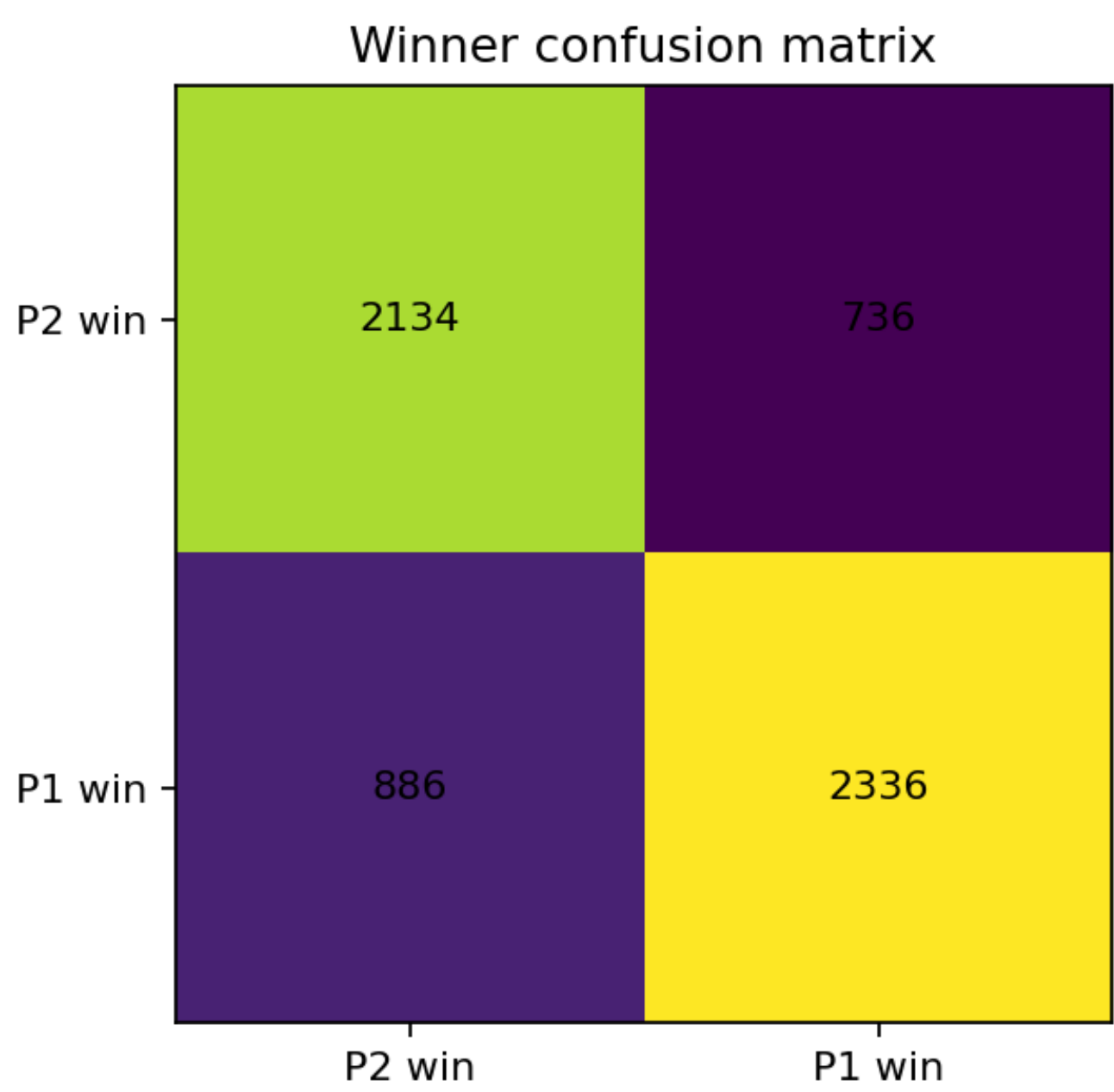
Learning rate:

$$\eta(g) = \eta_{\min} + (\eta_{\max} - \eta_{\min})e^{-g/\tau}$$

Time multiplier:

$$t_{\text{mult}} = 1 + c_s \left(1 - e^{-d/\tau_s}\right)$$

Winner Confusion Matrix



Final Predictive Model

The final system is a stacked gradient-boosted decision tree regression model trained separately for each sport.

For each sport:

- base differential model
- residual correction model
- total-points model

Final sport differential:

$$\hat{d} = b + 0.7r$$

This gave the best balance of accuracy, stability, and deployability.

Hyperparameters

Base differential model

- 900 iterations
- depth 5
- learning rate 0.03
- RMSE objective
- L_2 regularization = 6

Residual differential model

- 700 iterations
- depth 5
- learning rate 0.03
- RMSE objective
- L_2 regularization = 6

Total model

- 800 iterations
- depth 6
- learning rate 0.03
- MAE objective
- L_2 regularization = 6

Main Results Summary

Final performance:

- Match total-difference MAE: about 13.8
- Winner accuracy: about 73%

The final model predicts:

- sport-level score differences
- sport-level total points
- total match differential
- final winner

KD-Tree Confidence Insight

To connect the project to computational geometry, KD-trees were used to analyze local neighborhood structure in matchup feature space.

Key idea:

- dense historical neighborhoods $\rightarrow$ higher confidence
- sparse historical neighborhoods $\rightarrow$ lower confidence

This intuition works naturally for local models such as KNN, but was less effective for the final boosted-tree predictor.

Practical Output

The saved package supports:

- training on full historical data
- loading saved models
- predicting future matchups by player name
- returning sport-level and full-match outcomes

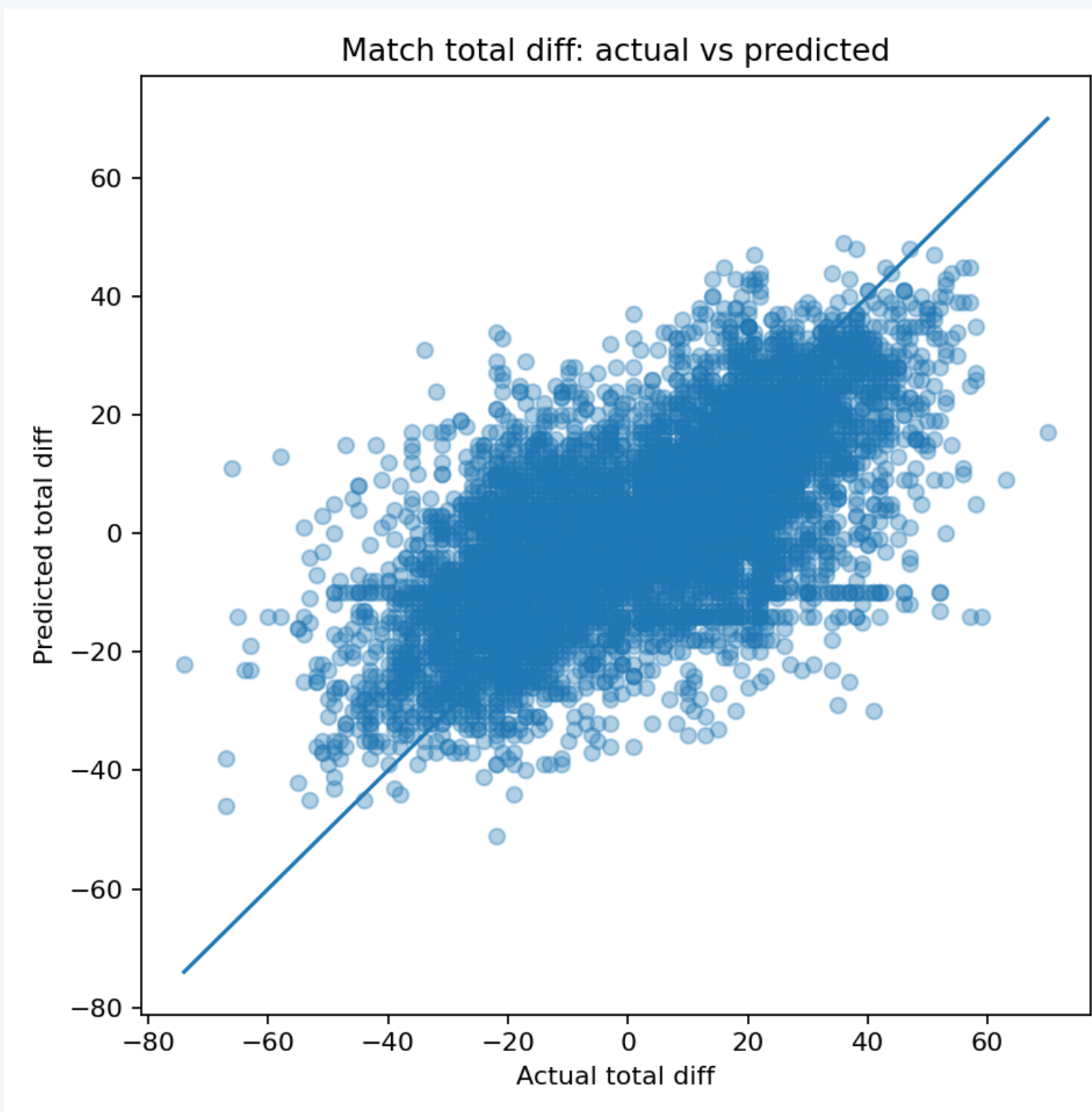
Conclusion

- Dynamic ratings and structured pre-match features are strong predictors of Racketlon outcomes.
- The final boosted-tree model gave the best balance of accuracy, stability, and deployability.
- KD-tree neighborhood analysis provided a computational-geometry interpretation of confidence.
- The complete system works end-to-end: scraping, cleaning, features, training, packaging, and future-match inference.

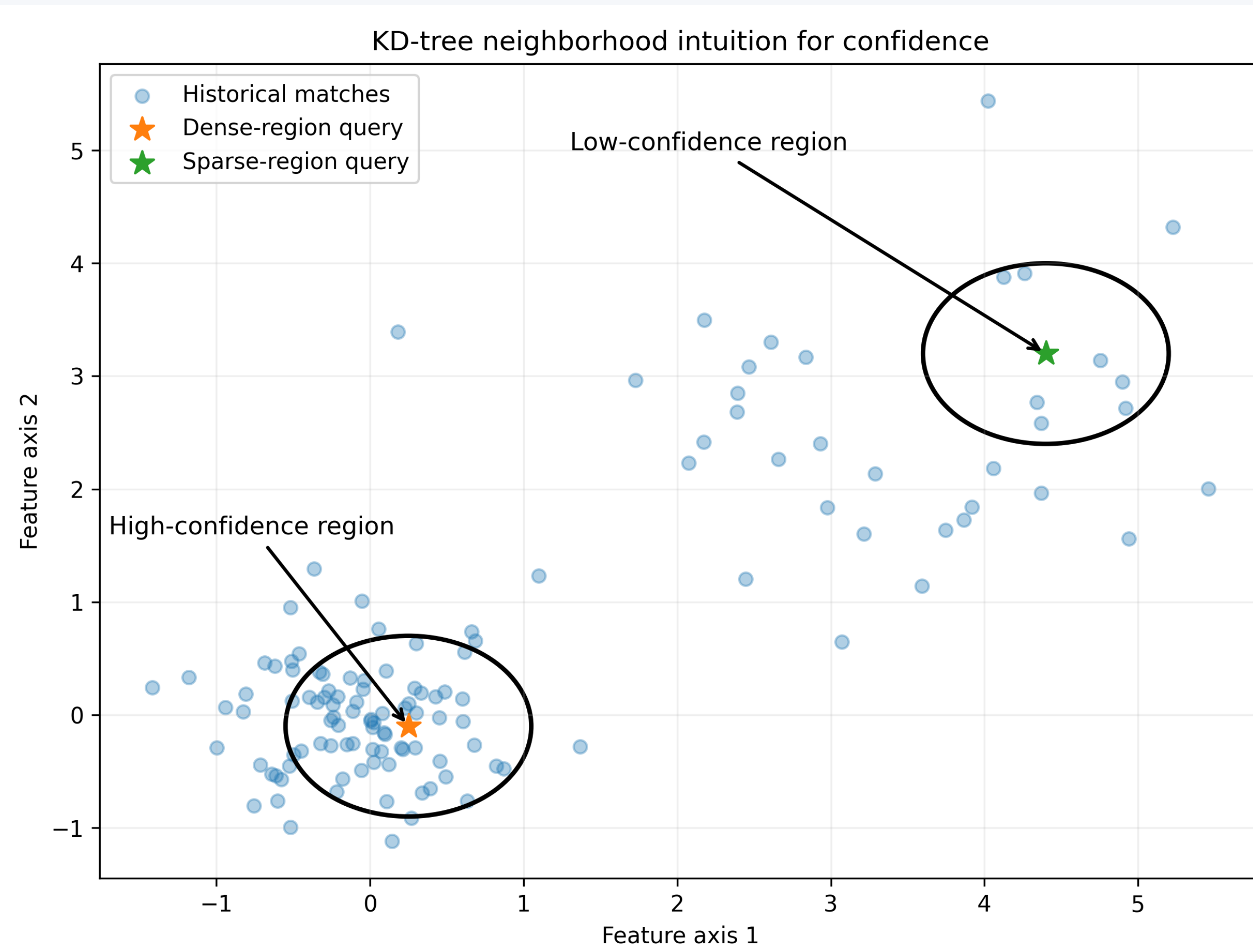
Actual Match

Semi final • Men A - Elite				
	Zain Magdon-Ismail		4	21 21 15
	Patrick Moran [2]		21	9 9 13
Sat 4/11/2026 1:15 PM 📍 Ainsworth Gym, Smith College				

Match Total Difference: Actual vs Predicted



KD-Tree Neighborhood Intuition



Model Prediction Output

Sample Model Output

Prediction Summary

MATCH PREDICTION

Zain Magdon-Ismail vs Patrick Moran

TT: 9 - 21 (diff -11.5)

BD: 21 - 12 (diff +8.6)

SQ: 21 - 15 (diff +4.6)

TN: 21 - 14 (diff +2.9)

TOTAL

Zain Magdon-Ismail : 72

Patrick Moran : 62

Total Diff: +10

Predicted Winner: Zain Magdon-Ismail

Player 1 Ratings

Zain Magdon-Ismail

TT: rating -18.62 pred -10.6 games 29

BD: rating 11.22 pred +8.8 games 29

SQ: rating 13.76 pred +8.2 games 29

TN: rating 7.18 pred +3.0 games 22

Player 2 Ratings

Patrick Moran

TT: rating 11.17 pred +6.8 games 67

BD: rating -1.08 pred -0.9 games 67

SQ: rating 8.56 pred +5.3 games 67

TN: rating 0.97 pred +0.4 games 53